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Using statistical process monitoring to identify us business cycle change points and turning points

David Enck[a], Mario Beruvides[a], Víctor G. Tercero-Gómez[b] and Alvaro E. Cordero-Franco[c]

[a]Department of Industrial Engineering, Texas Tech University Lubbock, United States; [b]School of Engineering and Sciences, Tecnologico De Monterrey, Mexico; [c]Universidad Autónoma De Nuevo León San Nicolás De Los Garza, Mexico

ABSTRACT

The Business Cycle paradigm of Mitchell and Burns has evolved from their original goal of understanding the entire economic process to the binary identification of growth and recessionary Turning Points. We propose a new paradigm for modelling the Business Cycle based on the Statistical Process Monitoring technique of Self-Starting Cumulative Sum (SSCUSUM) control charts. The SSCUSUM charts provide continuous characterization of aggregate economic activity through the identification of changes in the mean or standard deviation of economic indicators. A case study is conducted using real GDP % growth between 1965 and 2020 which shows that SSCUSUM charts: identify periods of steady state performance with statistically differentiable means and/or standard deviations, reliably reproduce the National Bureau of Economic Research Business Cycle Turning Points, identify patterns of economic activity leading up to and away from recessions, and identify twice the information on economic performance as the current bivariate approach. Over the study period, the SSCUSUM method identifies 42 changes in the mean or standard deviation of real GDP % growth, while the NBER TPs identify 8 peaks and 7 troughs.

KEYWORDS

Self starting cumulative sum control chart; Markov model; NBER business cycle dating committee; gross domestic product % growth

I. Introduction

Understanding a nations business cycle (BC) and recent economic status is critical to decisions made by its government, businesses, and citizens. Mitchell (1913; 1927a) looked to extend the view of a BC from focusing on extreme crises to focusing on the entire economic process of prosperity, recession, depression, and revival. As early as 1913 Mitchell realized that the business cycle was a complex dynamic process that required an end to end assessment of each unique BC, 'Every business cycle, strictly speaking, is a unique series of events and has a unique explanation, because it is the outgrowth of a preceding series of events, likewise unique (Mitchell 1913 pg. 450).' To that end, Burns and Mitchell (1946) created an approach to model this idea by summarizing and evaluating a large number of economic indicators across the entire economy; however, the method was simplified over time to the present-day approach used by the NBER which only requires identification of BC Turning Points (TPs) for peaks and troughs (NBER 2003a, 2008, 2016).

Research into the identification of BC TPs has focused on developing models to recreate the BC TPs as defined by the NBER. Performance is measured by how well a model can recreate the NBER BC TPs with the goal of providing transparency, consistency, and a reduced time to signal (TTS) the TPs. This current bivariate approach does not meet the original intent of Mitchell (1913; 1927a).

While business cycle Turning Point methods focused on historical and real-time identification of peaks and troughs, Business Cycle theory, regarding forecasting of economic performance, evolved through differing philosophical views on what drives an economy. These philosophies led to varying assumptions regarding the underlying Data Generating Process (DGP) for an economy, which resulted in a progression from: the Keynesian model (Keynes 1935) to initial microfoundations tied to Keynesian models (Clower 1979) to Real Business Cycles (Kydland and Prescott 1982; Long and Plosser 1983; Prescott 1986) and then to Dynamic Stochastic Generalized Equilibrium (DSGE) models

CONTACT David Enck david.enck@ttu.edu Department of Industrial Engineering, Texas Tech University Lubbock, United States

(Rotemberg and Woodford 1997; Smets and Wouters 2002). After the DSGE models failed to predict or provide a warning of the impending 2008 recession, the economics community and their stakeholders reflected on the failure. Congressional hearings held, by the Committee on Science and Technology, in 2010 (Congress, U S 2010) discussed the inability of these models to identify the economic downturn. As one of the 5 expert witnesses to testify, Dr. Winter commented that "Historical perspective is particularly important . . . we need more models that seek to capture systematic behavioural tendencies as they are . . ." rather than jumping to a complex model-based result. Another expert witness, Dr. Page made the point ". . . [for] a complex system, a single model can only cast so much light", while making the case for using multiple models when studying economic performance. This has led to a good deal of research into the performance of DSGE models and, more generally, what comes next for macroeconomic models. From reviewing the pros and cons of DSGE models (Krugman 2009; Blanchard 2016a, 2016b; Romer 2016; Stiglitz 2018) to proposing different models (Blanchard 2017a, 2017b, 2018; Wren-Lewis 2018), to developing approaches for addressing improvement opportunities related to assumptions ranging from rational expectations to mathematical issues such as estimation methods (Segnon et al. 2018; Dosi and Roventini 2019; Morelli et al. 2020; Tom 2020). We only provide a brief overview of the theoretical macroeconomics models because our proposal represents a precursor to these modelling approaches. Regarding Business Cycle theory, this paper presents a method that can help researchers understand real GDP growth and business cycle behaviour independent of an assumed DGP as was suggested by Dr. Winter in the Congressional hearings in 2010 (Congress, U S 2010). We recommend using Statistical Process Monitoring to model economic performance to gain an understanding of whether real GDP growth consists of periods of steady state performance in the mean or standard deviation and using these periods of steady state performance to evaluate the ability of more complex forecasting models to predict the Change Points. We will structure our paper in the terms of BC TPs; however, our third specified benefit relates to the ability to idenitfy CPs in the mean and standard deviation that can be used to test theory based BC forecasting models.

This paper proposes the use of the Statistical Process Monitoring (SPM) method of Self-Starting Cumulative Sum (SSCUSUM) control charts (Hawkins 1987) to identify historical and real time Change Points (CPs) in the mean and standard deviation of real Gross Domestic Product (GDP) % quarterly growth as a way of understanding and describing the aggregate economic activity across the entire economic process while also identifying BC TPs in an algorithmically rigorous manner.

The SSCUSUM charts partition real GDP % growth into segments of consistent statistical performance relative to the segment average, and standard deviation. The segment averages and standard deviations are used to represent how the economy is behaving across the entire business process. Given the statistically identified segments, patterns can be analysed, and rules can be applied to describe the BC including the declaration of TPs.

The current bi-variate characterization of economic performance implies that all recessions and growth periods are the same and that there is nothing to learn by studying the behaviour of economic performance within these states. This argument is troubling on its face as the economy is a dynamic process and recessions can be created by varying endogenous and exogenous forces which may cause different and identifiable patterns in the performance of economic indicators. The performance of key economic indicators leading up to a recession may be driven by a degrading economy or change may come about by a catastrophic external event, in each case the expected behaviour of the indicators preceding the recession would be different. Clearly an assessment of economic activity across the BC should provide the TPs as well as the behaviour of economic activity before and after these markers. The behaviour is not available with the bi-variety methods.

When comparing the SSCUSUM CP results to the NBER TPs, it is important to know that the NBER utilizes monthly and quarterly data on more indices than just real GDP to declare TPs. The NBER considers real GDP, real Gross Domestic

Income (GDI), economy-wide employment, real income, industrial production, and wholesale-retail sales, however their focus is on real GDP (NBER 2003bb). Research into historical identification of BC TPs has covered a range of methods from: qualitative univariate assessments utilized by the NBER (Bry and Boschan 1971) to more advanced methods that are meant to reliably recreate the BC TPs using reproducible statistical algorithms. The statistical algorithms have included univariate and multivariable methods: ARIMA (Hess and Iwata 1997), Markov Switching models (Hamilton 1989; Goodwin 1993; Chauvet and Hamilton 2005; Chauvet and Piger 2008; Hamilton 2011), logistic/probit models (Birchenhall et al. 1999; Pelaez 2005), and Machine Learning (Giustoa and Piger 2017). These methods utilize a univariate response; however, some of the multivariable methods include leading indicators or auto-regressive terms as regressors. The authors Golosnoy and Hogrefe (2009) further compared the capability of dynamic factor Markov switching models of Chauvet (1998) and multivariate CUSUM control charts of Healy (1987) for identifying TPs for peaks and troughs.

The SSCUSUM will be shown to have performance that is as good or better, relative to accuracy and timeliness, than the bivariate methods of Conditional Probability (Hamilton 2011) and the Markov Switching model (Chauvet and Piger 2003). The Conditional Probability model uses the real GDP % growth as a response with no additional leading indicators. The Markov Switching model includes a single auto-regressive term to help predict the BC TPs. Beyond the goal of identifying BC TPs there are three potential benefits of using SSCUSUM charts over bivariate methods. The first benefit is the ability to identify patterns in the performance of key indicators (e.g. real GDP % growth) as seen through the SSCUSUM CPs. The bottom chart of Figure 1 shows a specific example of a degrading pattern in the average of US real GDP quarterly growth that starts in 1983 and leads up to the NBER declared recession in 1990-Q3.

The bi-variate measurement of the current BC model focuses the investigation of economic performance on the start of the recession in 1990 while providing no indication of the 2 significant changes in aggregate economic activity that should be investigated and may be connected to the recession that occurred in 1990.

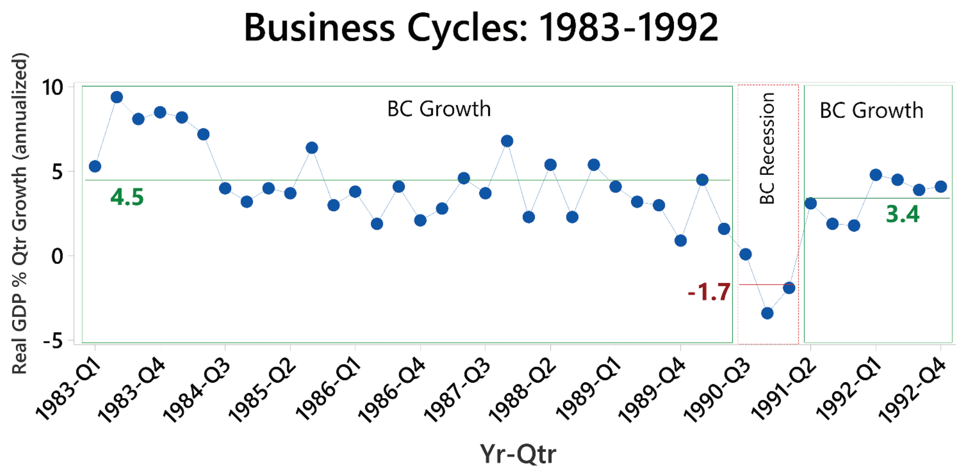
The second benefit relates to the ability to characterize model performance and reliability. Model performance is quantified using the Average Run Length (ARL). The ARL values can be selected based on the context of the analysis and established through the selection of the SSCUSUM model parameters, h and k, where k is an allowance for the minimal detectable change and h is a decision limit. The bivariate methods do not include these model performance characterizations. ARL_0 represents the average number of time-periods before the SSCUSUM chart indicates a change has occurred to the underlying system mean and/or standard deviation when no change occurred. ARL_0 is a measure of false alarms. ARL_1 represents the average number of time-periods until an out-of-control signal, if the underlying system mean and/or standard deviation have changed. ARL_1 is a measure of the sensitivity of the analysis to identify changes. A residual analysis can be used to determine whether the model is a reliable representation of the observed data.

The third benefit of using SSCUSUM charts stems from the statistically identified library of CPs that provide researchers a collection of testing opportunities to assess how well theoretical models predict actual changes in economic performance.

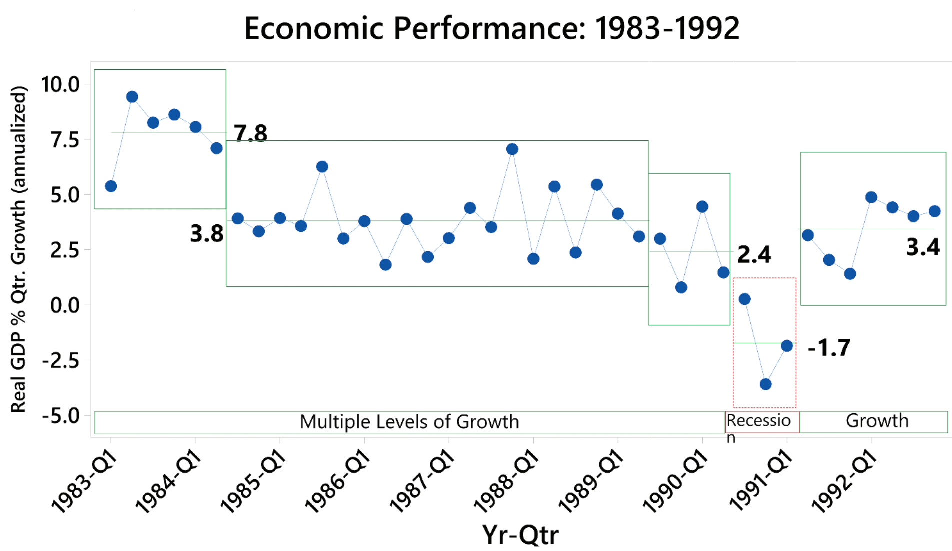
The next section presents a new Change Point definition for business cycles. Section III provides an overview of different Statistical Process Monitoring methods and criteria for selecting SSCUSUM charts. Section IV defines the SSCUSUM methods to identify Change Points in real GDP % growth. Section V presents a case study using SSCUSUM methods to quantify aggregate economic activity and identify BC turning points using a real-time data set. Section VI provides a summary of our findings and future research opportunities.

II. New Change Point Definition of Business Cycles

SSCUSUM Change Points (CPs) are more informative than BC Turning Points (TPs). While TPs represent the change between recessionary and



NBER breakdown of business cycle



CPs for economic performance identified with SSCUSUM chart

Figure 1. NBER BCDC declares 1983 to 1990 to be a single growth period. Periods of growth are assumed to be consistent which is clearly not the case. Bottom graph used the SSCUSUM method to identify different levels for the mean of real GDP % growth.

growth periods CPs represent shifts in the mean and/or standard deviation of the economic performance measure of interest. In this paper, we will use real GDP % quarterly growth.

The following population model for CPs accounts for changes in the mean and/or variation across sequences or segments of time for our key measure. Let Y_i be the real GDP % quarterly growth at time i. From a change-point perspective, Y_i is defined as:

$$Y_i \sim \begin{cases} N(\mu_0, \sigma_0), i = 1, \ldots, \tau_1 - 1 \\ N(\mu_1, \sigma_1), i = \tau_1, \ldots, \tau_2 - 1 \\ \vdots \\ N(\mu_k, \sigma_k), i = \tau_{k-1}, \ldots, \tau_k - 1 \end{cases} \tag{2.1}$$

where $\tau_1, \ldots, \tau_k$ corresponds to change-points with $(\mu_l, \sigma_l) : l = 1, \ldots, k$ as distribution parameters. Because the initial segment has a selected starting point rather than a CP, the subscript for the mean and standard deviation is set to 0.

The population model for CPs identifies a more comprehensive list of changes to real GDP % growth than the BC TPs. For the specific analysis of identifying BC TPs, for each CP a determination must be made as to whether the CP represents a BC TP. If it is decided that a CP represents a BC TP it will be called a SSCUSUM TP to differentiate it from the BC TP.

To define the rules for converting SSCUSUM CPs to SSCUSUM TPs one needs to understand the difference between how the SSCUSUM CP and NBER TP methods approach and quantify the problem of modelling variation in real GDP. The BC TP methods try to identify local optimums of real GDP, while the CP methods identify changes in the average or standard deviation of the % quarterly growth of real GDP.

An example of the differences between the two methods is described in Figure 2. The NBER TP date identified a local minimum even though that date was clearly part of the recovery in economic activity. For the goal of aligning with the NBER approach, the rule for converting a SSCUSUM CP to TP will follow the local minima/maxima

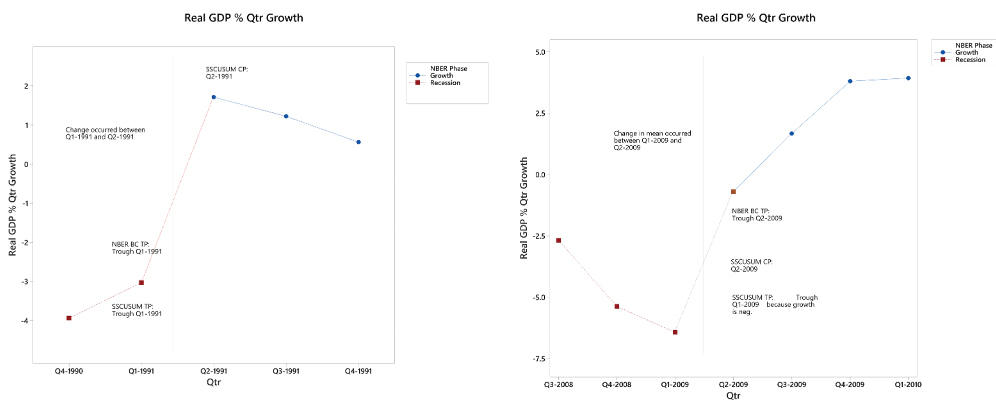


Figure 2. On left, the SSCUSUM CP started a segment with a larger positive mean % growth; therefore, the trough is defined to have occurred the previous quarter. Graph on the right, the SSCUSUM CP started a segment with a larger mean % growth; however, the growth was negative for that quarter, so trough is defined as starting that quarter.

1. For a CP that represents a decrease in the mean of real GDP % growth, the following rules apply:
   i. If the CP quarter has negative growth the peak SSCUSUM TP is defined as the previous quarter,
   ii. If the CP quarter has positive growth and the next quarter has negative growth the peak SSCUSUM TP is defined as the current quarter
2. For a CP that represents an increase in the mean of real GDP % growth, the following rules apply:
   i. If the CP quarter has negative growth the trough SSCUSUM TP is defined as the current quarter,
   ii. If the CP quarter has positive growth the trough SSCUSUM TP is defined as the previous quarter

Figure 3. Rules for converting SSCUSUM CPs to SSCUSUM TPs.

convention when the identified SSCUSUM CP is negative. An analogous example can be created for reductions in the mean or standard deviation of % growth in real GDP. The rules for converting SSCUSUM CPs to TPs are defined in Figure 3 and are used in the case study in Section V.

III. Statistical Process Monitoring Methods to Assess Economic Performance

To identify Business Cycle Change Points using real GDP % growth we propose the use of an SPM method developed by Hawkins (1987), SSCUSUM Control Charts for the mean and/or standard deviation. Table 1 describes the characteristics of several different types of control charts that are potential candidates for modelling economic performance.

The SSCUSUM chart performs well for small sustained shifts in the mean or standard deviation when there is limited data to establish new control limits after a shift has occurred. These characteristics describe the modelling environment for quarterly real GDP % growth.

Table 1. Comparison of control charts that may be used to monitor real GDP % growth.

Chart	Type of Change to Detect	Comments
Short run Shewhart charts (Quesenberry 1991)	Large shifts in mean/std. dev.	Reduced number of data points for Phase I study. Unusual observations will have an undue influence on the variation.
CUSUM (Page 1961)	Small sustained shifts in mean.	Requires Phase I study.
Exponentially Weighted Moving Average (Srivastava 1993)	Small sustained shifts in mean and/or std. dev.	Similar performance to CUSUM
Likelihood ratio test: unknown parameter (Hawkins and Zamba 2005)	Sustained shifts in mean or std. dev.	Performance not as good as SSCUSUM and more complex algorithm. Larger ARL_1 for the mean when k <0.75, larger ARL_1 than CUSUM when variance increases.
Lepage non-parametric control chart (Ross and Adams 2012)	General changes to a distribution	Requires at least 20 observations to start analysis (SSCUSUM can start with 3)
SSCUSUM (D. Hawkins 1987)	Small sustained shifts in mean and/or std. dev.	Does not require extensive Phase I, has acceptable ARL_1 and is easy to implement. Early unusual observations, first 3 to 7 observations, can inflate the standard deviation.

IV. SSCUSUM Methodology

The SSCUSUM analysis partitions real GDP % growth into segments determined by their state of statistical control along with the segment average, and standard deviation. The average and standard deviation for statistically differentiated and in control, segments can be used to easily compare the state of the economy between segments. The calculations depend only on the data from the current segment, which allows the SSCUSUM analysis to account for the dynamic nature of the US economy.

The SSCUSUM method accumulates deviations from a moving average or variance until the test statistic for the mean or variation exceeds a control limit, h, signalling a statistically identifiable shift. Let $\bar{Y}_j$ represent the mean of the first j quarters starting at an initial point or previous CP. S_j represents the sum of squared deviations for the same j quarters. The statistics $\bar{Y}_j$ and S_j can be calculated recursively via:

$$\bar{Y}_j = \bar{Y}_{j-1} + \left(Y_j - \bar{Y}_{j-1}\right)/j \qquad 4.1$$

$$S_j = S_{j-1} + (j-1)\left(Y_j - \bar{Y}_{j-1}\right)^2/j \qquad 4.2$$

Hawkins (1969) showed that a sequence of independent t-statistics, with *j-2* degrees of freedom, could be generated by comparing the current observation at time j with the preceding observations, using the recursive equations 4.1 and 4.2,

$$T_j = a_j\left(Y_j - \bar{Y}_{j-1}\right)/\sqrt{S_{j-1}/(j-2)} \qquad 4.3$$

where, $a_j = \sqrt{(j-1)/j}$. The independence of the T_j statistics, $j = 1, \ldots, n$, allows for understanding and control over the ARL characteristics of the SSCUSUM chart; however, the dependence of the T_j statistics on changing degrees of freedom adds complexity to its use as an analytical tool. The dependence of the T_j statistic on degrees of freedom can be removed by re-standardizing T_j to a N (0,1) using the following relation:

$$U_j = \varphi^{-1}\left(F_t\left(T_j, j-2\right)\right) \qquad 4.4$$

where $F_t\left(T_j, j-2\right)$ is the cumulative distribution function for the t-distribution. This quantile transformation method was developed by Quesenberry (1991).

Hawkins (1981) used the re-standardized U_j to define a cumulative sum statistic, V_j, that changes as the standard deviation of Y_j changes,

$$V_j = \left(\sqrt{|U_j|} - 0.822\right)/0.349 \qquad 4.5$$

where V_j is approximately N(0,1) when the process is under control.

Using V_j and U_j, four SSCUSUM statistics are defined to monitor the mean and standard deviation for increasing or decreasing trends.

Starting values for four SSCUSUM statistics are: $L_0^+ = L_0^- = S_0^+ = S_0^- = 0$. The SSCUSUM statistics are defined as follows:

$$L_j^+ = \max\left(0, L_{j-1}^+ + U_j - k\right), \qquad 4.6$$

for an increase in the mean,

$$L_j^- = \min\left(0, L_{j-1}^- + U_j + k\right), \qquad 4.7$$

for adecrease in the mean,

$$S_j^+ = \max\left(0, S_{j-1}^+ + V_j - k\right), \qquad 4.8$$

for an increase in the std.dev.,

$$S_j^- = \min\left(0, S_{j-1}^- + V_j + k\right), \qquad 4.9$$

for adecrease in the std.dev.,

where k is a parameter defined such that $2k$ defines the shift that the chart detects with the best possible performance. If the absolute value of one of the four SSCUSUM statistics grows beyond a control limit, h, the process is considered out of control. The CP is then identified by stepping back through the preceding time periods for the statistic that flagged an out of control until the last non-zero value in the sequence. Values of h can be determined for specific values of k through Monte Carlo simulation or Markov Chains. Hawkins and Olwell (1998) use a Markov Chain approximation to generate ARL_0 values with their program getanyh.exe and recently developed a package in R titled CUSUM design (Hawkins, Olwell, and Wang 2020).

The SSCUSUM statistics can be monitored on a control chart as shown by the graph on the right of Figure 4. Note that the CUSUM value at 1989-Q3 is the last negative value prior to the signal in 1990-Q4 which identifies 1989-Q3 as the CP. The visual display of the summary statistics is not easily interpretable so our approach is to identify the CPs using the SSCUSUM statistics in equations 4.6 to 4.9 and overlay the CPs on a time series plot of the real GDP % growth values as seen in the graph on the left of Figure 4.

In order to effectively utilize SSCUSUM charts to understand the entire economic process of prosperity, recession, depression, and revival we follow a 3-step process: i) characterize real GDP growth using the SSCUSUM to identify segments of consistent economic performance and Change Points (CPs) in the mean and variation, ii) use BC TP rules to convert the CPs to BC TPs to identify the

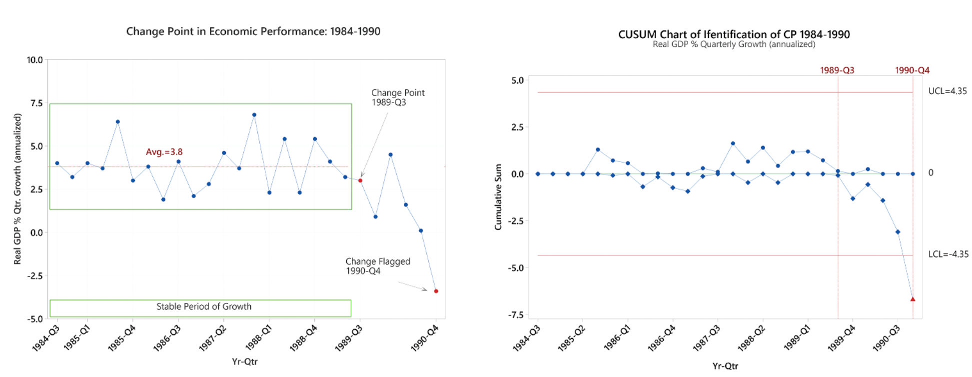


Figure 4. CUSUM chart for real GDP % quarterly growth (annualized), graph on the right. The period with a negative CUSUM prior to the 1990-Q4 flag, is the CP(1989-Q3). The graph on the left provides an easy to understand time series plot representing the CP.

1. Read vintage data file for real GDP growth: vintage are cols, quarters are rows.
2. Define initial conditions: use initial point as the first CP, CP_0.
3. While not the end of the final vintage:
   - 3.1 While not at the end of the current vintage:
     - 3.1.1 Search across quarters for a change in one of the four SSCUSUM statistics ($L^{+/-}$, $S^{+/-}$ from Section IV).
     - 3.1.2 If CP is flagged:
       - 3.1.2.1 Determine the new CP quarter, CP_{i+1}.
       - 3.1.2.2 Store the real GDP growth values between original CP_i and quarter prior to the new CP_{i+1}.
       - 3.1.2.3 Calculate the summary statistics between quarters of original CP_i and new CP_{i+1}.
       - 3.1.2.4 Store the quarter when CP_{i+1} was flagged, CP_{flag}
       - 3.1.2.5 Set current CP to quarter for CP_{i+1}
     - 3.1.3 If no CP is flagged:
       - 3.1.3.1 CP remains the same.
   - 3.2 Shift the vintage to the next available.
4. Use real GDP growth values and CP list to generate SSCUSUM graph.
5. Use Summary statistics to generate bar chart for real GDP growth.

Figure 5. Algorithm for the SSCUSUM analysis accounting for real-time data.

start of recessions and growth periods, and iii) analyse the CP patterns in growth periods and characterize their behaviour leading up to and away from recessions. As will be seen in Section V there are at least 2 types of behaviours for CPs leading up to and away from recessions.

V. Case Study

Introduction

This case study will assess the capability of the SSCUSUM model to: i) identify patterns in aggregate economic activity across the BC, ii) assess model reliability and performance, and iii) create a library of statistically identified CPs. The identification of patterns in aggregate economic activity consists of two parts, identification of BC TPs and the patterns in CPs leading up to and away from the TPs. In Section V the accuracy of the SSCUSUM TPs and average TTS was compared with the results of the: Conditional Probability model of Hamilton (2011) and the Markov Switching model of Chauvet and Piger (2003). The models use real GDP as their measure of economic activity. An evaluation of the ARL and residual analysis was conducted in Section V and 6.4.1, respectively. After the SSCUSUM CPs have been converted into TPs, we evaluate the patterns in aggregate economic activity created by the changing average of real GDP % growth leading up to and away from the 7 recessions between 1965 and 2020.

Real-Time Data Set

The case study uses real-time data for real GNP/GDP from 1965 to 2020. The data were transformed to % quarterly growth at a compounded annual rate of change (CARC). The real-time data set was downloaded from the Federal Reserve Bank of Philadelphia Real-Time Data Set for Macroeconomists (RTDSM) (FRED 2020).

Real GDP is available in vintages for each quarter and year starting in 1965. A vintage provides the real GDP that was available for all previous quarters at a point in time. There are 222 vintages between Q4-1965 and Q2-2020. Each vintage was available in a column labelled ROUTPUTyyQq. The yy and q represent the year and quarter the vintage was available.

The Philadelphia Fed publishes a series of comments indicating issues with vintage updates. One important note clarifies that real Gross National Product was used up until the end of 1991 when the measure was updated, real Gross Domestic Product was used until the end of 1995 when it was switched to a chain weight GDP starting in the first quarter of 1996. Prior to 1996 fixed weights had been used.

The percent change between quarters was calculated using a variant of the compound interest formula as provided by the BEA, $[(GDP_{(i+1)}/GDP_i)^{(4/1)} - 1]*100$ where 4 represents the periodicity, based on 4 Quarters, with 1 representing the time between periods (BEA 2017).

A confirmation test was conducted on the calculation of the real GDP percent change, based on data from the RTDSM (FRED 2020) as compared to data reported by the BEA. The RTDSM reported data using the advance report for 2018-Q3, while the BEA used the second revision to the 2018-Q3 data. Both sites had data available from 1970-Q1 to 2018-Q3. The confirmation test was conducted on a vintage of data collected prior to the final analysis.

The calculated percent change using the RTDSM data set matched all quarters as

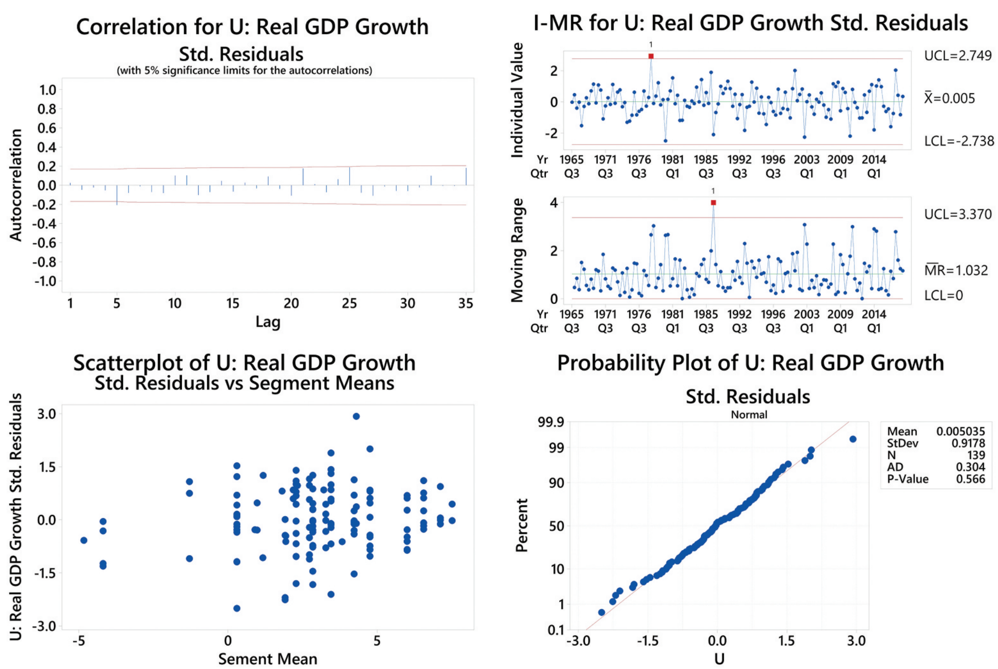


Figure 6. Residual analysis of real GDP % quarterly growth (annualized) covering 1965 to 2020. The residuals were standardized using the within sequence mean and standard deviation in a manner analogous to weighted least squares (Kutner, 2005).

Table 2. For each k, the table contains original h values for the first CP of an in-control (IC) process and h values from a Monte Carlo simulation to detect the second CP in the same direction. 10,000 runs per ARL_0. Orig. h represents the original h values and Adj. h represents the adjusted h values.

	k									
	0.10		0.25		0.50		0.75		1.00	
ARL_0	Orig. h	Adj. h	Orig. h	Adj. h	Orig. h	Adj. h	Orig. h	Adj. h	Orig. h	Adj. h
50	*4.567*	*3.473*	*3.340*	*2.540*	*2.225*	*1.700*	*1.601*	*1.179*	*1.181*	*0.832*
100	*6.361*	*4.791*	*4.418*	*3.480*	*2.849*	*2.300*	*2.037*	*1.646*	*1.532*	*1.232*
200	*8.520*	*6.510*	*5.597*	*4.560*	*3.502*	*2.970*	*2.481*	*2.167*	*1.874*	*1.620*
370	*10.722*	*8.390*	*6.708*	*5.665*	*4.095*	*3.630*	*2.882*	*2.604*	*2.175*	*1.979*
500	*11.890*	*9.490*	*7.267*	*6.245*	*4.389*	*3.940*	*3.080*	*2.826*	*2.323*	*2.153*

Table 3. Summary statistics are available for each Quarter. This table begins at 1965-Q1, CP_0 and displays the data used in the identification of the 1966-Q1 CP, (CP_1), which was a reduction in mean.

Yr-Qtr	Obs	Avg	Avg_seq	U	SSCUSUM Loc minus	SSCUSUM Loc plus	h Loc plus/ minus	V	SSCUSUM Scale minus	SSCUSUM Scale plus	h Scaleplus/ minus
1965-Q1	9.23	9.23	7.51	0.00	0.00	0.00	NA	0.00	NA	NA	NA
1965-Q2	5.09	7.16	7.51	0.00	0.00	0.00	NA	0.00	NA	NA	NA
1965-Q3	7.02	7.11	7.51	9.02	0.00	0.00	3.34	−1.92	−1.67	0.00	4.42
1965-Q4	8.68	7.51	7.51	0.44	0.00	0.19	3.34	−0.44	−1.87	0.00	4.42
1966-Q1	6.03	7.21	NA	−0.52	−0.27	0.00	3.34	−0.28	−1.90	0.00	4.42
1966-Q2	1.89	6.32	NA	−1.83	−1.85	0.00	3.34	1.52	−0.13	1.27	4.42
1966-Q3	4.04	6.00	NA	−0.64	−2.24	0.00	3.34	−0.07	0.00	0.95	4.42
1966-Q4	4.57	5.82	NA	−0.43	−2.42	0.00	3.34	−0.47	−0.22	0.23	4.42
1967-Q1	0.00	5.17	NA	−1.74	−3.91	0.00	3.34	1.42	0.00	1.40	4.42

Table 4. Sensitivity study to assess the ARL_0 values and k with respect to how close the SSCUSUM CPs are to the BCDC TPs. The ARL_0 values with a * next to them used the modified h values specified in Table 2. The ARL_0 = 17 with the modified h values provide the best performance relative to averages and individual observations.

				Sensitivity Analysis			
ARL_0	ARL_0: Loc ±	ARL_0: Scale ±	k	Avg #Qtr. before Peak	# Peaks Missed	Avg #Qtr. before Trough	# Troughs Missed
13	50	50	0.25	−0.17	1	0.50	1
17	50	100	0.25	−1.17	1	0.67	1
17	50*	100*	0.25	−0.50	1	0.50	1
17	50	100	0.5	0.17	1	0.83	2
17	50*	100*	0.5	1.00	1	0.83	1
25	100	100	0.25	−0.83	1	−0.33	1
25	100*	100*	0.25	−0.83	1	0.17	1
25	100	100	0.5	−0.83	1	−0.20	2
25	100*	100*	0.5	−0.17	1	−0.17	1

reported by the BEA except for 2018-Q3 which was 3.4 vs. 3.5%. The difference was expected given the different vintages reported. The calculated values are used in this case study.

Methods

The algorithm for the SSCUSUM analysis, accounting for real-time data, is provided in Figure 5, Figure 6. Table 2 provides adjusted h values that can be used as control limits for real-time monitoring of real GDP % growth.

Table 3 shows the summary statistics that are created as the algorithm is executed. The CP_1 of 1966-Q1 was flagged in 1967-Q1 (CP_{flag}). The SSCUSUM location (loc) reduction statistic was −3.91 in 1966-Q1 with a control limit (h value) of −3.34. The CP_1 was identified by tracing the Loc minus statistic back from 1967-Q1 to the last negative value, 1966-Q1. As can be seen in Table 3, the segment from 1965-Q1 to 1965-Q4 has a mean of 7.51% growth. Table 3 shows the status of the table prior to restarting the analysis at 1966-Q1. The restart process resets all the summary statistics starting at 1966-Q1 just as it had at 1965-Q1. The average and standard deviation for the previous segment was stored as the last result prior to the identified CP.

Table 3 demonstrated how the output of the SSCUSUM algorithm identifies CPs and stores summary statistics. To generate Table 3, the h and k values need to be defined. The settings for ARL_0 for the average, as well as the value for k were determined via a sensitivity study displayed in Table 4.

The sensitivity study conducts a relative comparison between different ARL_0 settings to understand the sensitivity of SSCUSUM methods when applied to real GDP % growth data. ARL_1 represents the ability to detect specific shifts in a process mean. We use the # of quarters before a TP is detected as a surrogate for assessing ARL_1. The smaller the average # of quarters the lower the ARL for the shift in process average for the TPs.

All SSCUSUM charts in the sensitivity study missed, at a minimum, 1 trough and 1 peak during the two recessions in the early 1980s. The SSCUSUM charts combined these into one longer recession. Our analysis drops these BC TPs that were missed as the deviation from the CPs cannot be calculated. The study found that the SSCUSUM charts with ARL_0 set to 50 for the average and 100 for the standard deviation with k = 0.25 and using the modified h values, from Table 4, gave average CPs that were close to the TPs with an ARL_0 of 17 quarters or 4.25 years. Given the performance statistics were similar to the ARL_0 = 13 chart the improvement in the ARL_0 of 4.25 years over 3.25 was considered significant when the study spanned close to 45 years. The full analysis resulted in 42 CPs vs 15 TPs so there was no need to pick an ARL_0 less than 17.

Sensitivity studies will not always be needed. Given this was the first application of SSCUSUM to the analysis of real GDP % growth a sensitivity study was in order, future work can pick h and k values based on characteristics of the ARL values and behaviours observed in this study.

Analysis

The analysis of the patterns in aggregate economic activity using SSCUSUM follows a 3-step approach. The first 2 steps statistically identify CPs using SSCUSUM and the 3rd step applies the business rules that convert objectively identifiable statistical patterns to defined business patterns. The SSCUSUM 3-step process can be defined as:

(1) Fit the model, and check residuals,
(2) Identify CPs, generate the CP chart, and summary statistics for each quarter using real-time data, and
(3) Convert SSCUSUM CPs to SSCUSUM TPs and compare to BC TPs.

Fit the model and check residuals

The model fit is represented by the mean and standard deviation for each segment of in control data. Using the methodology described in Sec. IV each quarter adds new data to the summary statistics, including standardized residuals which are represented as the column U in Table 3. The standardized residuals are assessed to determine if they meet the model assumptions: i) a mean of zero, ii) equal variance within a segment, iii) little or no autocorrelation, and iv) fit a Normal distribution. As can be seen from the SSCUSUM equations in Section 4, the calculation of standardized residuals can only start after the second observation, at the earliest. The first two observations of each segment will always be 0 and are therefore not included in the residual analysis.

The residual analysis in Figure 6 shows that, for the standardized residuals across all segments, the mean is zero, there is no practical autocorrelation, the variance is consistent within a segment, and the residuals could have come from a Normal distribution.

The key to a proper residual analysis for SPM is to conduct a within sequence analysis of residuals. Investigators often conduct the residual analysis on the original observations and identify significant

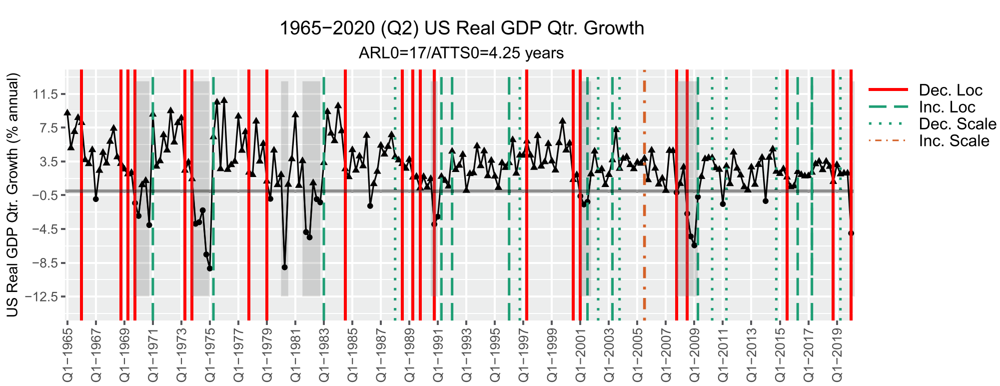


Figure 7. Business cycle CPs for real GDP % quarterly growth. The Q2-2020 observation of −31.7% was dropped to avoid skewing the graph. The SSCUSUM that generated the CPs used an ARL_0 =50 for location and ARL_0 =100 for scale. The h values are listed in Table 2 for k= 0.25. Grey boxes represent recessions and the type of CPs are specified in the legend. Positive values are represented by a triangle while values below zero are dots.

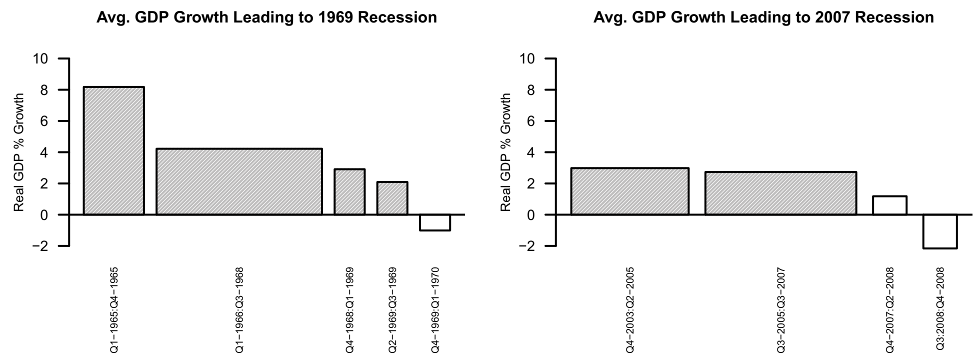


Figure 8. Pattern on the left seen in 5 of the 7 recessions between 1965 and 2020, degradation pattern with 1 to 2 CPs within a 2-year period of the BC TP. Pattern on the right seen in 3 recessions where the CP and TP occur simultaneously, or the TP was not identified. The non-shaded bars include the NBER declared recession.

autocorrelation; however, this result is driven from changes in the mean over time rather than autocorrelation between consecutive quarters.

Identify CPs, generate the CP chart, and summary statistics

The statistical program R, was used to analyse the data from Q1-1965 to Q2-2020, identify the CP library, and generate Figure 7. Displaying results for a SSCUSUM analysis usually takes the form of a control chart for the SSCUSUM statistics described in Section IV. We follow the standard practice for BC analysis and display the real GDP % growth in Figure 7. The CPs are represented by different types of vertical lines, which are defined in the legend.

As discussed in the Introduction, Figure 8 shows there are differences in the pattern of economic activity leading up to and away from the recessions. The following patterns can be seen in the graph:

- recessions that were led by degrading economic performance:
  - for 5 of the 8 recessions at least 1 CP that occurred within the year prior to the recession and,
  - for 2 of the 5 recessions with 1 CP within 1 year, there were 2 CPs within 2 years prior to the recession,

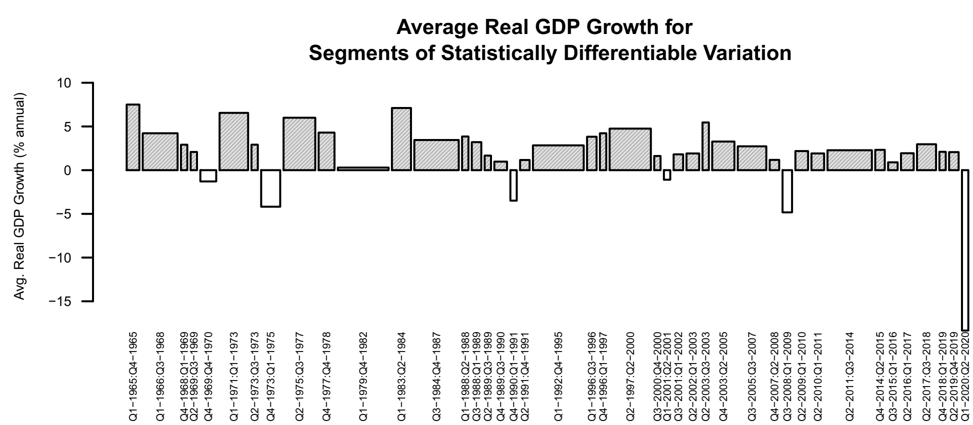


Figure 9. Segment means that were statistically identified by the SSCUSUM analysis show the 42 segments of quarterly data for real GDP % growth. White bars represent SSCUSUM segments that include NBER declared recessions.

- recessions with sudden shifts in economic performance:
  - 2 CPs occurred on or 1 quarter after the start of the recession and
  - 1 trough and 1 peak were not identified, they were included within the Q1-1979 to Q2-1983 CPs that signalled an overall low average for that segment. These TPs were obfuscated by sudden shifts in real GDP.

Observationally, there are at least 2 categories of recessions that can be observed from this analysis. The recessions that have 1 or 2 CPs prior are different from the recessions where the CP and TP occurred together with no prior CP for location within 1 year. The 2 types of recessions occurred before and after 1984 so they appear to be independent of the Great Moderation (Stock and Watson 2002). Figure 8 provides a visual example of the two types of recessions that were observed in the case study.

There is also evidence for 2 types of growth periods coming out of a recession. Between 1965 and 1984 the US exited 4 recessions and within 2 quarters of the TP the growth was above 8%. After 1984 the US exited 3 recessions with growth approximately 3%, 2% and 2%. The 2 types of growth exit patterns are confounded with the Great Moderation so we will leave them for future evaluation. These potential patterns leading up to and away from recessions provide an opportunity for researches to classify recessions and test the ability of economic models to predict patterns of behaviour in economic activity.

Figure 9 presents a simplification of the SSCUSUM chart in Figure 7. The figure presents the averages for the segments with identified differences in mean or standard deviation. This simple representation shows the changes in economic performance that can be uncovered by identifying continuous changes in economic activity rather than focusing strictly on the binary measure of growth and recessionary periods as represented by the NBER focused models.

This type of information allows for simple comparisons and identification of changes in economic activity based on statistically identified segments rather than ad hoc comparisons such as year over year or current quarter to the same quarter last year.

Comparison of owners with non-owners

Convert SSCUSUM CPs to SSCUSUM TPs and compare to BC TPs

This section will apply the rules for converting SSCUSUM CPs to SSCUSUM TPs as developed in Section II and compare the SSCUSUM TPs with the BC TPs on the criteria of accuracy and TTS for each BC TP. The Conditional Prob. and Markov Switching Model mentioned in Section 1, were included in the comparison.

All models in this comparison used real-time data sets. The SSCUSUM, Conditional Prob., and Markov models use real GDP % growth as their key performance indicator. The main difference between the models includes how historical data are used and how the signals for change of state are defined. The Conditional Probability and Markov Switching models used all the data from the study to estimate the mean for recession and growth periods along with the standard deviation which was assumed constant. The SSCUSUM method identified each CP using the current segment of data only, which enables the ability to model the dynamic nature of the US economy.

All 3 models used a historical assessment of real-time data to define the cut-off points which define a change of state. The Conditional Probability model outputs have been kept up to date by Hamilton (Hamilton and James 2010). The Markov Chain model (Chauvet and Piger 2008) results are available up to 2001.

Table 5 provides an overview of the results of the three models as compared to the announcement dates and BC TPs of the NBER. Inclusion of the 2008 and 2020 recessions for the SSCUSUM and Conditional Probability models changed the averages by an average of 0.2 quarters with a max. change of 0.47 quarters so the comparison vs. Markov models was not severely impacted.

Averages start at Q1-1980, they do not include the Q3-1980 and Q3-1981 NBER dates or the

Table 5. Representation for the TPs and average TTS for the NBER, SSCUSUM, Conditional Prob., and Markov Switching models. The first 4 NBER TPs have NA for announcement dates because the NBER did not announce TPs until 1979. Averages for SSCUSUM do not include the Q3-1980 and Q3-1981 NBER dates. Averages for all 3 methods start at Q4-1969. # of Qtrs. until Flag, for all models, starts at the NBER TP.

NBER				SSCUSUM				Cond. Prob.			Markov Switching		
NBER TP	**Peak/ Trough**	**NBER Announcement Date**	**# of Qtrs until Announcement**	**SSCUSUM CP**	**# of Qtrs Until Flag**	**SSCUSUM TP**	**Accuracy rel. NBER**	**Cond. Prob. TP**	**# of Qtrs Until Flag**	**Accuracy rel. NBER**	**MSM TP**	**# of Qtrs Until Flag**	**Accuracy rel. NBER**
Q4-1969 0	Peak	NA	NA	Q4-1969	1	Q3-1969	−1	Q4-1969	2	0	Q4-	1969	1
Q4-1970 0	Trough	NA	NA	Q1-1971	5	Q4-1970	0	Q1-1971	3	1	Q4-	1970	1
Q4-1973 1	Peak	NA	NA	Q4-1973	1	Q3-1973	−1	Q4-1973	2	0	Q1-	1974	1
Q1-1975 1	Trough	NA	NA	Q2-1975	3	Q1-1975	0	Q2-1975	4	1	Q2-	1975	2
Q1-1980 1	Peak	Q2-1980	1	Q1-1979	1	Q1-1979	−4	Q2-1979	−1	−3	Q2-	1980	2
Q3-1980 0	Trough	Q3-1981	4	No CP	-		-	Q4-1980	3	1	Q3-	1980	1
Q3-1981 0	Peak	Q1-1982	2	No CP	-		-	Q2-1981	2	−1	Q3-	1981	1
Q4-1982 1	Trough	Q3-1983	3	Q1-1983	4	Q4-1982	0	Q1-1983	3	1	Q1-	1983	2
Q3-1990 1	Peak	Q2-1991	3	Q4-1990	2	Q3-1990	0	Q3-1990	2	0	Q4-	1990	2
Q1-1991 4	Trough	Q4-1992	7	Q2-1991	6	Q1-1991	0	Q3-1992	8	6	Q1-	1992	5
Q1-2001 2	Peak	Q4-2001	3	Q1-2001	2	Q1-2001	0	Q2-2001	4	1	Q3-	2001	3
Q4-2001 0	Trough	Q3-2003	7	Q3-2001	3	Q3-2001	−1	Q3-2001	3	−1	Q4-	2001	1
Q4-2007	Peak	Q4-2008	4	Q4-2007	4	Q3-2007	−1	Q3-2008	5	3			
Q2-2009	Trough	Q3-2010	5	Q2-2009	3	Q2-2009	0	Q3-2009	4	1			
Q4-2019	Peak	Q2-2020	2	Q1-2020	2	Q4-2019	0	Q4-2019	2.66	0			
			Avg TTS		Avg TTS		Avg TP Accuracy		Avg TTS	Avg. Accuracy		Avg TTS	Avg. Accuracy
Troughs:			5.2		4.0		−0.2		4.0	1.4		2.0	1.0
Peaks:			2.6		1.9		−1.0		2.3	0.0		1.7	0.8
Overall:			3.9		2.8		−0.6		3.1	0.7		1.8	0.9

SSCUSUM dates prior to Q1-1979. The SSCUSUM chart did not flag the Q3-1980 to Q3-1981 recession as special cause variation and the NBER did not make announcements prior to 1980.

# of Qtrs until Flag: starts at the NBER Business Cycle Turning Point

A comparison of the data available in Table 5 shows that the SSCUSUM method is a very capable tool for the identification of BC TPs. In the discussion section consideration will be given to the fact that the SSCUSUM analysis missed the 2 TPs in the 1980s. Table 5 indicates the following:

(1) all models have a shorter TTS than the NBER,
(2) algorithms identify TPs within 1 quarter of the NBER TP date: 92%, 83%, and 80% of the time for: SSCUSUM, Markov Model, and Cond. Prob. respectively,
(3) the Markov Model has an 50% larger deviation from NBER TPs and 36% shorter TTS than the SSCUSUM method,
(4) the Conditional Probability model has a 17% larger deviation from NBER TPs and 11% longer TTS than the SSCUSUM,
(5) starting with the Q4-1969 TP, the SSCUSUM, Markov and Cond. Prob. models all missed one TP by more than 4 quarters or more,
    a. the SSCUSUM missed the Q3-1980 trough, which led to missing the Q3-1981 peak. If the Q2-1980 value had been slightly lower, the Q3-1980 Trough would have been flagged and the Q3-1981 would have been detected,
    b. the Cond. Prob. model suffered a similar issue by declaring the start of the Q1-1991 Trough as 6 quarters later than the NBER TP due to a close call on the decision criteria, and
    c. the Markov model missed the Q1-1991 trough by declaring the TP as 1 year later than the NBER.

Discussion

The case study has revealed the benefits and some opportunities when using SSCUSUM to identify patterns and BC TPs in aggregate economic activity. Our case study revealed at least 2 types of CP behaviours leading to recessions and identified 42 CPs overall vs 15 BC TPs identified by the bi-variate methods.

The SSCUSUM method is similar in accuracy and timeliness as compared to the Conditional Probability and Markov Models, however it is clearly not as sensitive to large short-duration spikes as methods that are designed to assess a bi-variate response and use all historical information in modelling. The fact that the SSCUSUM method missed the Q3-1980 trough and Q3-1981 peak is a significant deviation from the NBER announced dates and deserves comment. As discussed in Section III, the SSCUSUM is designed to identify small sustained shifts in the mean and/or standard deviation. The residual analysis showed that the SSCUSUM model fit the data, reinforcing that the missed identification of 2 BC TPs reflected the design goal of the chart. This issue can be addressed by using SPM charts designed to catch this type of shift. To confirm this approach, a Q-chart (Quesenberry 1991) analysis was conducted and was able to identify the Q3-1980 trough. Modifications to approaches that combine CUSUM and Shewhart control charts (Shewhart 1925 Lucas 1982; Ottenstreuer, Weiss, and Knoth 2019) could be explored to adapt the methodology to SSCUSUM charts and Q-Charts. The Q3-1981 peak was missed because the previous trough was not identified. The increase in the variation impacted the ability of the SSCUSUM to identify the peak. If the Q-Chart had been used to identify the Q2-1980 trough, the SSCUSUM analysis would have identified the Q3-1981 CP by Q4-1981, which matches the Markov Switching model.

Additional opportunities for future research include leveraging historical data and modelling segments that contain trends. Pooling estimates of variation from previous segments can potentially increase the degrees of freedom when standardizing each observation and re-standardizing for the tests in equations 4.3 and 4.4. The SSCUSUM methodology is able to properly identify these segments and they do not impact the method's ability to identify CPs or BC TPs, however there is an opportunity to test for and represent the transition patterns with linear models to provide a better description of the segment behaviour.

VI. Conclusion and Future Work

Understanding aggregate economic activity of historical data depends on the model and assumptions used when reducing the observations into an understandable pattern. Using the SSCUSUM approach we identified statistical evidence that, as an open dynamic system, the US economy shifts between segments of steady state or common cause growth and recessionary segments. A new point of view arises when viewing the economy systemically, thus using this perspective begs the question that policy makers must ask becomes, 'is the system in-control (i.e. in steady state)'? Reacting to a system as out of control when it is not brings the policy maker to the age-old reality of trying to fix something that is not broken, while ignoring a system that has changed, clouds their understanding of the danger on the horizon.

SPM offers a new paradigm for the modelling and analysis of aggregate economic activity. SSCUSUM performs well at identifying BC TPs, additionally we have shown that the CPs can help researchers and analysts define patterns across the business cycle that can be used to evaluate economic models and theories.

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ORCID

David Enck http://orcid.org/0000-0003-0289-3466
Mario Beruvides http://orcid.org/0000-0003-4579-4556
Víctor G. Tercero-Gómez http://orcid.org/0000-0002-5196-3451
Alvaro E. Cordero-Franco http://orcid.org/0000-0003-4389-7682

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